

Exoplanet Classifier: A Written Explanation

by Adhrit Chakraborty, for the Celesta Exoplanet Challenge (NASA Kepler data)

1. How I explored and cleaned the data

The data comes from NASA's Kepler space telescope. It is a table of about 9,564 possible planet signals, and each one is described by roughly 140 measurements. My first step was simply to get to know it. Three things stood out. The groups are uneven, with about half the signals being false alarms, a little under a third being confirmed planets, and only about a fifth still waiting as candidates. Many of the numbers, like how long a planet takes to orbit or how much its star dims, stretch across an enormous range. And a few columns are almost completely empty. Cleaning the data is where I made my most important decision. A few columns quietly give the answer away, because they hold conclusions that NASA's own checking process had already reached. The clearest example is the planet's name, since only confirmed planets are ever given one. Columns like that would make my results look incredible, but the model would only be copying, and it would fall apart on a brand new signal. So I throw those columns out inside the model itself, and I wrote a test that fails if even one of them slips back in. After that I removed empty and repetitive columns, filled the small gaps with sensible averages, put everything on a common scale, and was left with 91 honest measurements that are all rooted in real physics.

2. Why I chose the model I did

I did not want to just guess, so I trained twelve different models on the same data and compared them fairly. A family of models that learn in small steps, known as gradient boosting, came out clearly on top. They are very good at picking up the tangled patterns in a table like this, and they need very little fiddling. I went with one called HistGradientBoosting because it scored highest, trained quickly, and handled missing values gracefully. To be honest, the best few models were so close that picking a winner was almost a coin toss. I then fine tuned the model with a tool called Optuna, and I added one twist that I am proud of. Instead of only chasing the highest score, I asked the search to also penalise any model that did much better on practice data than on fresh data. In plain terms, I rewarded models that genuinely learn instead of ones that simply memorise. And because the challenge is really a yes or no question, is this a real planet or not, I also built two simpler models that each answer a single yes or no, sitting alongside the main one that sorts signals into three groups.

3. What I found, and what the model learned

The main model gets things right about 85% of the time on a balanced scoring measure, with very strong overall separation between the groups. The simpler real planet versus false alarm model passes 90% on both the practice and the unseen test data, and a tighter confirmed versus false alarm version reaches about 96%. All of this is achieved without cheating. The part I like most is what the model pays attention to. When I looked inside, the clues it leans on are the same ones a real astronomer would use: how strong and clean the dip in starlight is, how long the dip lasts compared with the orbit, whether the star's light shifts to one side during the dip (a hint that the signal is really coming from a nearby star), and how large the object would have to be. Its mistakes make sense too. Nearly all of the confusion is between candidate and confirmed, which is fair, because a candidate is simply a planet we have not confirmed yet. I also measured what my honesty costs: if I had used those forbidden answer columns, my score would have jumped by about 9%. I chose to leave that on the table.

4. Explaining it to someone without a science background

Picture a camera watching a streetlight. Every now and then something passes in front and dims it for a moment. Was it a person, or just a moth near the lens? My model is the detective that decides, except it does this for stars and planets. For every signal it asks a few simple questions. How deep and clean was the dip in brightness? Did the star wobble, the way it would if another star, rather than a planet, were tugging on it? Is the light truly coming from the target star, or leaking in from a neighbour? How big would the object have to be? From the answers it gives a clear verdict of real planet, maybe, or false alarm, along with how confident it is, such as 92%. Best of all, it shows its working. For every prediction it draws a small chart of which clues pushed it one way or the other, so anyone can follow the reasoning rather than just trust a label. In everyday terms it is a sorting tool. It points scientists toward the signals most worth a closer look through a real telescope, and quietly sets aside the obvious false alarms. And because it reasons from real evidence instead of copying an answer key, you can trust it on something it has never seen before.